

An Automatic Following/Forming Framework for Inverter-Based Resources

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Nomenclature

Symbols

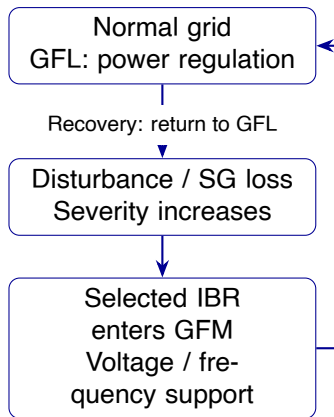
D_v, M	VSG damping / virtual inertia constant		
E	GFM internal voltage magnitude reference	u	exogenous inputs: set-points, loads, events
E'_q, E'_d	SG transient internal EMFs (q/d -axis)	V_i, θ_i	bus i voltage magnitude / phase angle
E''_q, E''_d	SG subtransient internal EMFs (q/d -axis)	x	full differential state vector
F_0, F_1	IBR vector fields, GFL / GFM branch	x_{SG}, x_{IBR}	SG / IBR state vectors
f_0, f_{coi}, ω_b	nominal / centre-of-inertia frequency; $\omega_b = 2\pi f_0$	Y	network admittance matrix
$g = 0$	network current balance at every bus, PF and TS	y	algebraic variables: bus voltages
I_{dc}, V_{dc}	converter DC-link current / voltage	δ, ω	SG rotor angle / speed
i_d, i_q	converter filter currents (d/q -axis)	θ_{PLL}, ξ_{PLL}	PLL estimated angle / integrator state
k_p, k_i	PI proportional / integral gains	$\theta_{VSG}, \omega_{VSG}$	VSG internal angle / angular speed
m_i	IBR i mode: 0 = GFL, 1 = GFM	ξ_P, ξ_Q	power-loop integrator states
P_j, Q_j	net active / reactive injection at bus j	ξ_{Id}^b, ξ_{Iq}^b	current-loop integrators, $b \in \{GFL, GFM\}$
t	time	ξ_{Vd}, ξ_{Vq}	voltage-loop integrator states

Abbreviations

EMF	electromotive force
GFL / GFM	grid-following / grid-forming control
IBR	inverter-based resource
PF / TS	power flow / time-domain simulation
PLL	phase-locked loop
SG	synchronous generator
VSG	virtual synchronous generator

Introduction

- Increasing inverter-based resource (IBR) penetration.
- Less synchronous-machine inertia.
- Weak grids challenge PLL-based grid-following (GFL) control.
- Grid-forming (GFM) control provides voltage and frequency support.
- Adapt the control mode to grid conditions [1],[2].



System Description and Operating Principles

Power flow

Rotor and converter states are constant; bus currents balance.

$$0 = g(x, y, u_0, m). \quad (1)$$

$$P_i^{inj} - P_j(x, y, u_0, m) = 0, \quad (2)$$

$$Q_i^{inj} - Q_j(x, y, u_0, m) = 0. \quad (3)$$

Time-domain simulation

Machine and converter states respond to disturbances.

$$\dot{x} = f(x, y, u, m, t), \quad (4)$$

$$0 = g(x, y, u, m, t), \quad (5)$$

$$m_i = \begin{cases} 0 & \text{GFL,} \\ 1 & \text{GFM.} \end{cases} \quad (6)$$

$$x = [x_{SG}^T, x_{IBR,1}^T, \dots, x_{IBR,4}^T]^T, \quad (7)$$

$$x_{SG} = [\delta, \omega, E'_q, E'_d, E''_q, E''_d]^T. \quad (8)$$

Shared physical states

$$x_p = [i_d, i_q, V_{dc}, I_{dc}]^T \quad (9)$$

$$\begin{aligned} \frac{L_f}{\omega_b} \dot{i}_d &= v_{cd} - v_d - R_f i_d \\ &+ \omega_{pu} L_f i_q, \end{aligned} \quad (10)$$

$$\begin{aligned} \frac{L_f}{\omega_b} \dot{i}_q &= v_{cq} - v_q - R_f i_q \\ &- \omega_{pu} L_f i_d. \end{aligned} \quad (11)$$

$$\omega_{pu} = \begin{cases} 1 + \dot{\theta}_{PLL}/\omega_b & m_i = 0, \\ \omega_{VSG} & m_i = 1. \end{cases} \quad (12)$$



Figure 1: IBR electrical model.

$$\dot{x}_{IBR} = \begin{cases} F_0(x_{IBR}, y, u) & m_i = 0, \\ F_1(x_{IBR}, y, u) & m_i = 1. \end{cases} \quad (13)$$

Switchable GFL/GFM Control Model

Controller state vectors

$$x_{GFL} = [\theta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{Id}^{GFL}, \xi_{Iq}^{GFL}]^T \quad (14)$$

$$x_{GFM} = [\theta_{VSG}, \omega_{VSG}, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}^{GFM}, \xi_{Iq}^{GFM}]^T \quad (15)$$

GFL: PLL + power control

$$\dot{\theta}_{PLL} = k_p, PLL v_q + k_i, PLL \xi_{PLL} \quad (16)$$

GFM: VSG + voltage control

$$M\dot{\omega}_{VSG} = \kappa P_{ref} - P_{inv} - D_v(\omega_{VSG} - 1) \quad (17)$$

$$\kappa = S_{base}/M_{base} \quad (18)$$

$$x_{IBR} = [i_d, i_q, V_{dc}, x_{GFL}^T, x_{GFM}^T, I_{dc}]^T \quad (19)$$

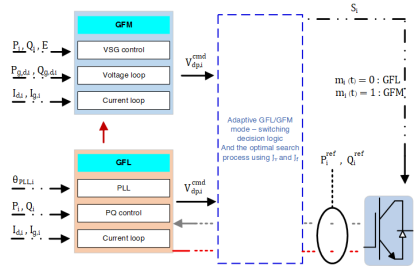


Figure 2: Switchable controller branches.

Mode Switching: Supervisory Readiness

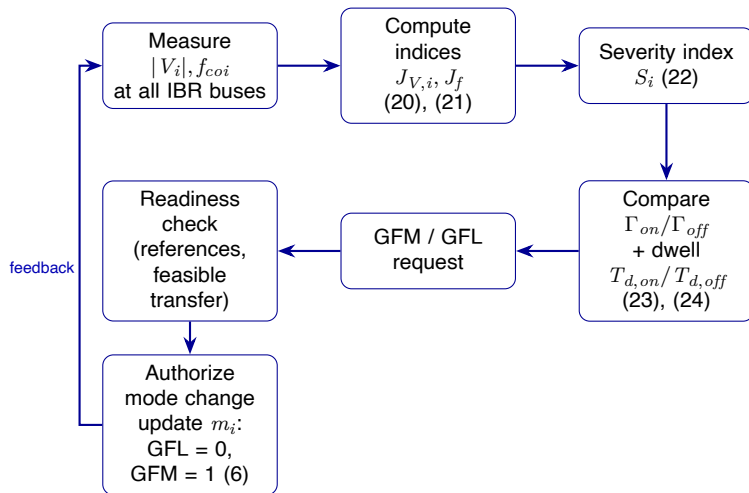


Figure 3: Supervisory readiness and GFL/GFM mode-switching procedure.

Mode Switching: Index and Logic

Voltage and frequency indices

$$J_{V,i} = \frac{||V_i| - V_i^*|}{\Delta V_{base}}, \quad (20)$$

$$J_f = \frac{|f_{coi} - f_0|}{\Delta f_{base}}, \quad (21)$$

$$S_i = \text{sat}_{[0,1]}(w_V J_{V,i} + w_f J_f). \quad (22)$$

$J_{V,i}, J_f$: voltage / centre-of-inertia indices; $S_i \in [0, 1]$: severity; V_i^* : healthy per-bus PF magnitude; $\Gamma_{on}, \Gamma_{off}$: thresholds; $T_{d,on}, T_{d,off}$: dwells; w_V, w_f : weights; $\mathcal{L}, \mathcal{G}$: online GFL/GFM sets.

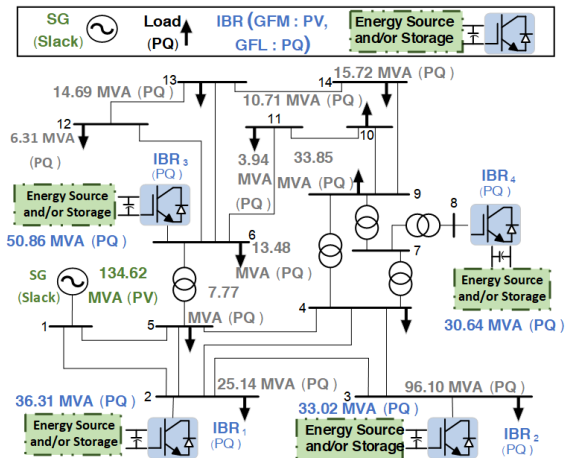
$$\text{GFM request: } \exists i \in \mathcal{L} : S_i \geq \Gamma_{on} \quad \text{for } T_{d,on} \quad (23)$$

$$\text{GFL request: } \forall i \in \mathcal{L} \cup \mathcal{G} : S_i < \Gamma_{off} \quad \text{for } T_{d,off} \quad (24)$$

Γ_{on}/Γ_{off} : 0.65/0.35; dwell: 0.10/1.00 s.

Index bases: 0.10 pu, 0.50 Hz; voltage/frequency weights: 0.5/0.5.

Case Study: IEEE 14-Bus System



SG1: Bus 1

IBR₁: Bus 2

IBR₂: Bus 3

IBR₃: Bus 6

IBR₄: Bus 8

100 MVA

60 Hz

Figure 4: IEEE 14-bus study network [3].

Parameter Settings: Network and Plant

Parameter	Symbol	Value	Unit
Network and synchronous machine			
System bases	S_{base}, f_0	100,60	MVA, Hz
SG (system base)	H, D	2.50,1.00	s, pu
SG time constants	$T'_{d0}, T''_{d0}, T''_{q0}$	7.40,0.03,0.033	s
IBR plant			
Filter	L_f, R_f	0.15,0.015	pu
DC bus / current limit	$C_{dc}, V_{dc}^0, I_{max}$	0.10,1.00,1.20	pu
DC source	$\epsilon_{dc}, P_r, \tau_s, \zeta^*$	0.10,1.00,0.005,0.707	–, pu, s, –
Chopper	$V_{dc}^{max}, \delta_{ch}$	1.10,0.02	pu, –

* Units follow the normalized controller equations.

Parameter Settings: Control and Supervisor

Parameter	Symbol	Value	Unit
GFL control			
PLL PI gains	$k_{p,PLL}, k_{i,PLL}$	1.20, 5.00	*
Power PI gains	$k_{p,P/Q}, k_{i,P/Q}$	0.80, 2.50	*
Shared current loop			
Current PI	$k_{p,I}, k_{i,I}$	0.30, 4.00	*
GFM control			
VSG	M, D_v	0.08, 20.0	pu [†]
Q–V loop	k_Q, k_E, τ_E	0.25, 8.00, 0.05	*, *, s
Voltage PI	$k_{p,V}, k_{i,V}$	1.20, 4.50	*
Supervisor			
Hysteresis	$\Gamma_{on}, \Gamma_{off}$	0.65, 0.35	–
Dwell / weights	$T_{d,on/off}, w_V, w_f$	0.10/1.00; 0.5, .50	s; –
Hold / lock	T_{hold}, T_{lock}	1.000, 2.000	s
Index bases	$\Delta V_{base}, \Delta f_{base}$	0.10, 0.50	pu, Hz

* Units follow the normalized controller equations.

† Report notation: $M = 2H_v$, $H_v = 0.04$ s.

Scenario Timeline

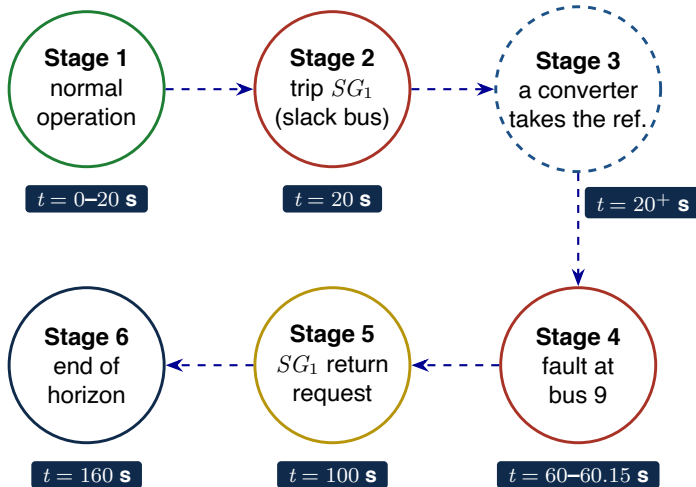


Figure 5: Scenario timeline. Breaker reclose after synchronism: 102.363 s; final IBR return to GFL: 116.238 s.

Result: IBR State Response

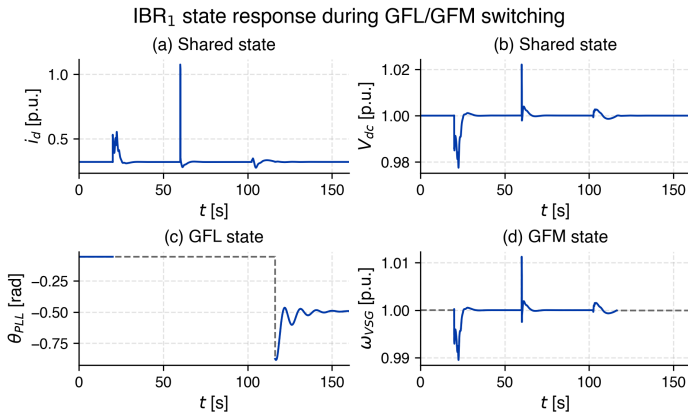


Figure 6: IBR₁ states: (a) i_d, i_q ; (b) V_{dc}, I_{dc} ; (c) ω_{VSG} ; (d) E

Blue = active; grey dashed = held stored state. IBR₁ only.

Result: Severity Index and Converter Mode

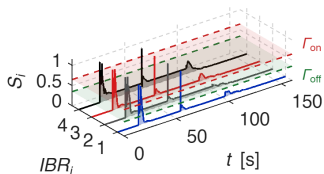


Figure 7: Severity index S_i per converter.

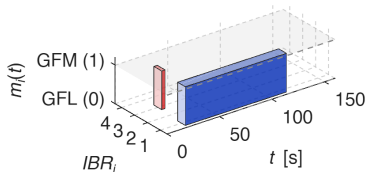


Figure 8: Converter mode m_i per converter: GFL = 0, GFM = 1.

Result: Voltage and Frequency

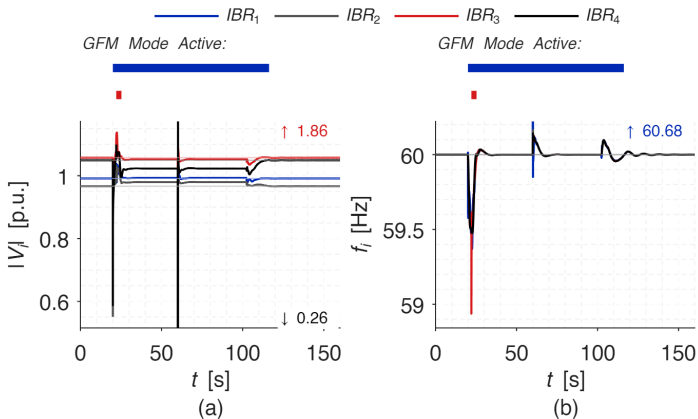


Figure 9: Converter responses: (a) bus-voltage magnitude and (b) converter frequency.

Result: Active and Reactive Power

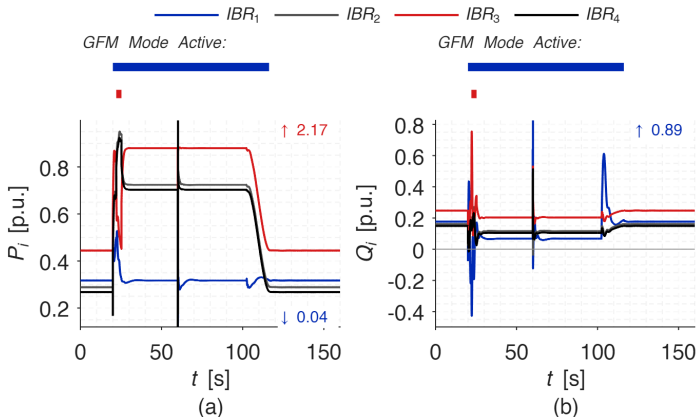


Figure 10: Converter power responses: (a) active power and (b) reactive power.

Fault Bus Voltage and Frequency

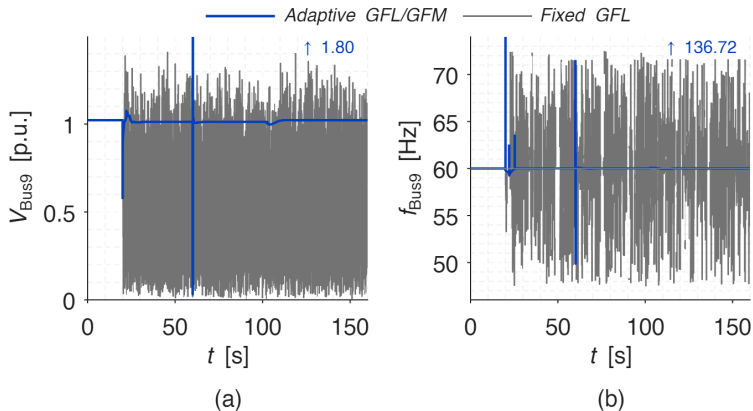


Figure 11: Bus-9 responses for adaptive GFL/GFM and fixed-GFL cases: (a) voltage magnitude and (b) frequency.

Fault Bus Active and Reactive Power

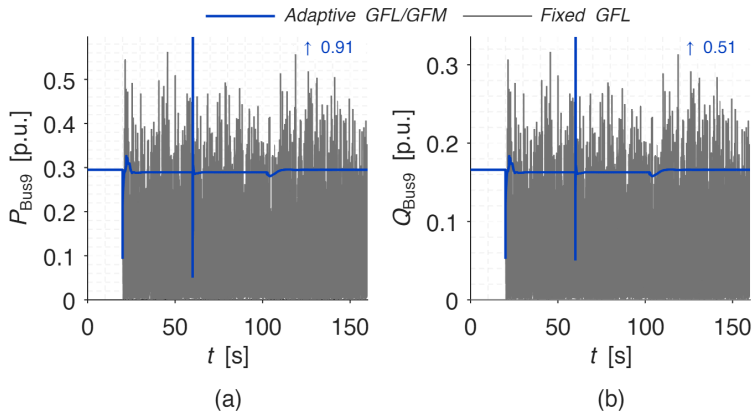


Figure 12: Bus-9 power responses for adaptive GFL/GFM and fixed-GFL cases: (a) active power and (b) reactive power.

Completed Development

- ☒ Power flow and time-domain simulation.
- ☒ Switchable GFL/GFM converter model.
- ☒ Continuous physical and DC states during mode transfer.
- ☒ Adaptive switching based on voltage and frequency deviations.

Future Development

- ☐ Establish admissible switching and dwell-time conditions for the hybrid DAE.
- ☐ Simulation on larger-scale power systems.
- ☐ Development of Intelligent switching control to manage transients and reduce switching errors

References

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